

Quantitative Spatial Models for Transport Appraisal

Optional capstone add-on: Berlin's U5 connection

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Draft skeleton

Policy question

- ▶ How should we **appraise a transport investment when households, firms, wages, rents, employment, and location choices respond?**
 - ▶ Conventional appraisal emphasizes time savings and user benefits.
 - ▶ A quantitative spatial model **adds endogenous sorting and general-equilibrium adjustment** to conventional appraisal.
- ▶ Highly topical
 - ▶ Allen, Fuchs, and Wong, *From Social Savings to Spatial Equilibrium*, RSUE (forthcoming) for further reading.
 - ▶ Currently working on related implementation work with the UK Department for Transport (with Sidharth Moktan, and Daniel Sturm).
- ▶ This capstone applies that logic to a **concrete case combining the ARSW2015 toolkit and the TTMATRIX toolkit.**

A quantitative transport-appraisal workflow

1. Define two comparable transport networks.
 - ▶ Baseline network
 - ▶ Counterfactual policy network (usually with extension)
2. Compute bilateral travel-time matrices.
 - ▶ Map travel times into commuting costs (usually via iceberg cost in model).
3. Quantify a baseline spatial equilibrium.
 - ▶ Estimate or set all parameters
 - ▶ Invert fundamentals conditional on parameter values and observed data
4. Solve the counterfactual policy equilibrium.
 - ▶ Compute relative changes from baseline to counterfactual equilibrium, $\hat{x} = \frac{x'}{x}$
 - ▶ Compute local changes in prices and quantities
 - ▶ Compute aggregate changes in welfare and population.

Running example: Berlin's central U5 connection



- ▶ Berlin's most recent U-Bahn extensions created a new east–west connection through the historic centre:
 - ▶ **2009:** Hauptbahnhof–Brandenburger Tor opened as the U55.
 - ▶ **2020:** Brandenburger Tor–Alexanderplatz opened, closing the gap and integrating the U55 into the U5.
- ▶ Our counterfactual adds **both new sections** to generate a sizable accessibility shocks

Network construction

Baseline network (2005 perspective)

- ▶ Take current Berlin line and station layers
- ▶ Delete Alexanderplatz–Hauptbahnhof connection
 - ▶ Copy the line layer in e.g. QGIS
 - ▶ Delete all corridor segments between the endpoints
 - ▶ Preserve intersecting services
 - ▶ Treat observed stations consistently

Counterfactual network

- ▶ Current network $\equiv$ future network from 2005 perspective
- ▶ Counterfactual network differs from baseline network by adding U5 Alexanderplatz–Hauptbahnhof extension

From GIS to bilateral travel times

- ▶ For transport counterfactuals, **need to compute travel times them ourselves**
 - ▶ Historic or future travel times can typically not be queried
 - ▶ Important that we use **exactly the same modelling and parameters**
- ▶ Travel times in ARSW2015 toolkit are fairly sophisticated (took us ages...)
 - ▶ U-Bahn, S-Bahn, Tram, Bus, interchange penalties, etc.
- ▶ **TTMATRIX toolkit designed for quick stylized counterfactuals.**
 - ▶ Just needs origins-destination shapefile, one network shapefile, 2 speed parameters
 - ▶ Station shapefile is optional (here, use is highly recommended).
 - ▶ Same blocks as in ARSW2015 toolkit; default speed parameters applicable.
 - ▶ Line shape corresponds to *counterfactual*. You just need to create *baseline network*.
 - ▶ **Run TTMATRIX toolkit for *counterfactual* and baseline *baseline network***
 - ▶ Generate two travel time matrices.
 - ▶ **Only change the network**, keep everything else (including parameters) the same.

Figure 2: direct travel-time effects

Direct accessibility effects of the central U5 connection

A. Mean travel time



B. Blocks reachable within 30 minutes



Relative change = $100 \times (\text{with U5} / \text{without U5} - 1)$. The origin block itself is excluded; red line shows the U5 connection.

Matrix validation before economics

- ▶ **Same dimension and exact identifier ordering.**
 - ▶ Minutes, not seconds; documented diagonal convention.
 - ▶ No unexplained missing or infinite OD pairs.
 - ▶ Residence-by-workplace orientation consistent with model.
 - ▶ Here commuting costs are symmetric, so does not matter
- ▶ **Accessibility indices should make sense**
 - ▶ Higher accessibility in centre and near stations
 - ▶ For affected OD pairs, complete-network travel times should weakly improve relative to the reduced network.
 - ▶ There should be no sizable travel time increases.
 - ▶ Check (again) after reading into MATLAB (using MAPIT)

Bridge to the ARSW model

Let t_{ij} denote travel time from residence i to workplace j .

1. Ensure both TTMATRIX outputs and ARSW-toolkit block index are in same order.
 - ▶ Ensured by construction, toolkits are compatible ✓
2. Convert t_{ij} into the commuting-cost object used by the model.
 - ▶ Ensured via exponential iceberg transport cost ✓
3. Treat the reduced network as the baseline transport system.
 - ▶ Replace the original ARSW-toolkit with the baseline TTMATRIX.
4. Recover or re-invert baseline fundamentals under that system.
 - ▶ Keep the parameter values constant at values used/estimated by ARSW.
5. Load the ARSW-toolkit counterfactual matrix and solve the model
 - ▶ While holding the chosen fundamentals and parameters fixed.

Equilibrium outcomes

► Before starting with the simulations

- **Inspect the fundamentals** inverted under the new travel times
- Commuting market access is particularly relevant
- All looks good $\Rightarrow$ most likely no ordering issue with blocks ✓

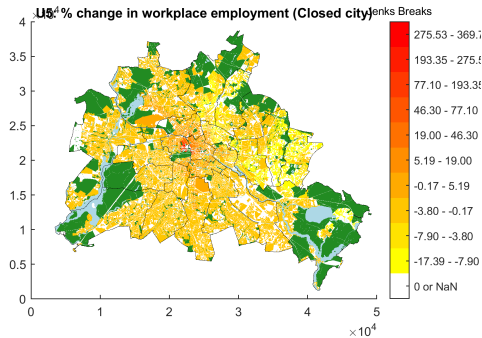
Inverted primitives

► Ready to simulate the model in the open-city and closed-city scenarios

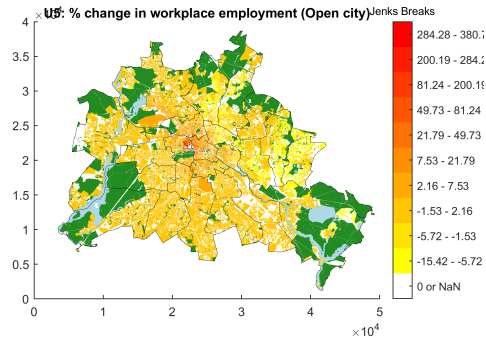
- Residence and workplace employment
- Wages and expected worker income
- Floor-space prices and development density
- Commuting flows and average commuting costs
- Productivity and amenity feedbacks under endogenous agglomeration
- Welfare or aggregate population

Employment response: closed versus open city

Closed city

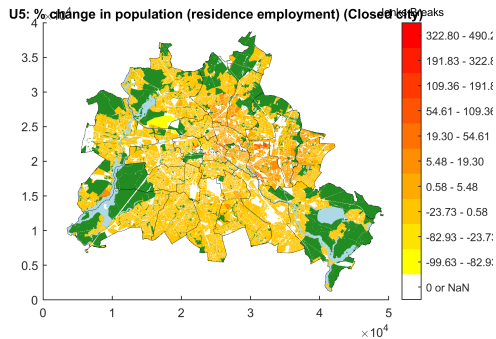


Open city

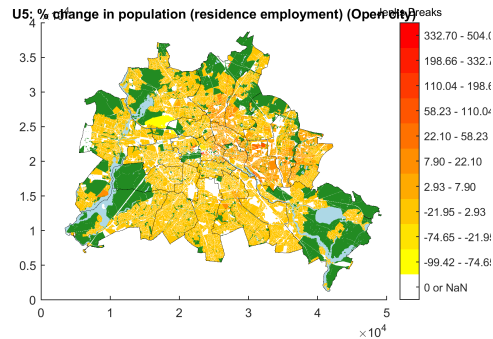


Population response: closed versus open city

Closed city

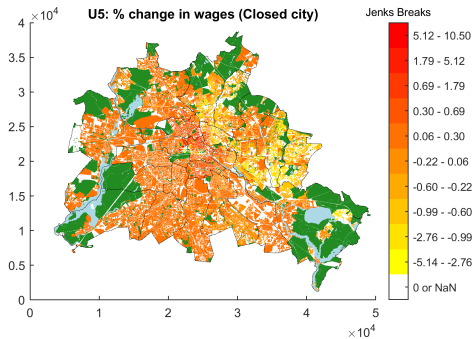


Open city

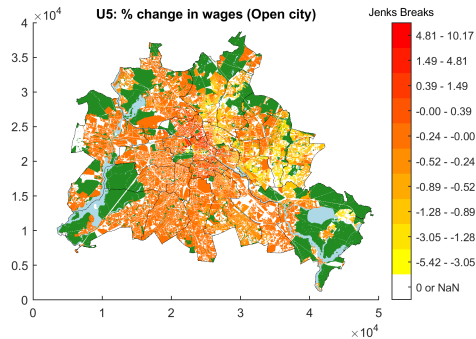


Wage response: closed versus open city

Closed city

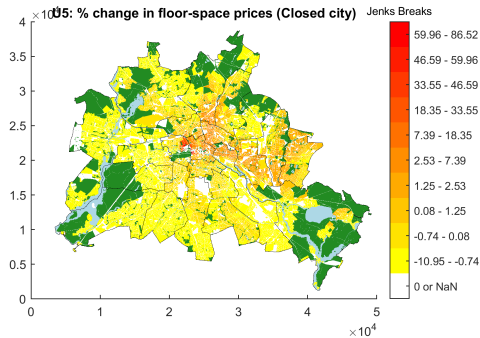


Open city

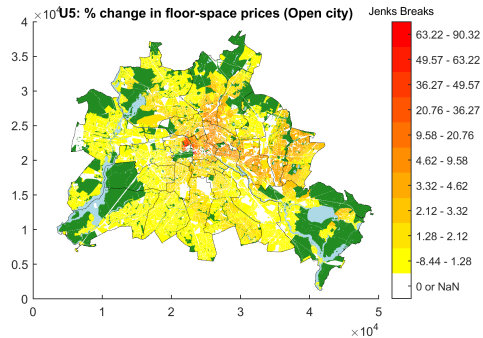


Floor-space price response: closed versus open city

Closed city



Open city



Aggregate effects depend on the spatial closure

Closure	Expected utility	Population	GDP	Land value
Closed city	+0.444%	0.000%	+0.307%	+1.229%
Open city	0.000%	+2.340%	+2.348%	+9.394%

- ▶ Land rent is the developer residual $r_i K_i = Q_i L_i - M_i$, not total floor-space rent $Q_i L_i$.
- ▶ The corrected land share is $1 - \mu = 0.25$. With fixed structures, M_i is fixed, so land rent absorbs the full change in floor-space revenue.
- ▶ Using the 2006 wage bill, baseline annual land rent is $0.25[(1 - \alpha)/\alpha + (1 - \beta)]W = 0.125W \simeq \text{EUR } 4.67\text{bn}$.
 - ▶ Implied annual land-rent gain: about **EUR 57m** closed city and **EUR 439m** open city.

At a common 5% capitalization rate, these annual gains correspond to approximately EUR 1.15bn and EUR 8.77bn. They measure incidence on landowners and are not added to the welfare benefit.

A rough annual monetary equivalent of the welfare gain

Back-of-the-envelope scaling

$$0.4439\% \times \underbrace{\text{EUR 37.347 bn}}_{\substack{\text{Berlin gross wage bill, 2006} \\ \text{excluding marginal employment}}} \simeq \text{EUR 166 million per year.}$$

- ▶ This computes the monetary equivalent of the closed-city welfare effect.
 - ▶ Recall that welfare scales linearly in wage
 - ▶ A recurring **annual** equivalent under the model's new spatial equilibrium.
- ▶ It does not yet deduct construction, operating or maintenance costs.
 - ▶ Notice that operating costs are about covered by fees

Source: Amt fuer Statistik Berlin-Brandenburg, VGR, P I 2, table 18. Model outcomes: authors' calculations.

Benefits and costs on a common 2006 basis

$$PV_{2006}(B) = \frac{\text{EUR } 166\text{m}}{0.05} \simeq \text{EUR } 3.32\text{bn.}$$

- ▶ Deflate both segments with the Destatis price index for engineering structures:
 - ▶ U55: EUR 320m in 2009 → EUR 288m in 2006 prices.
 - ▶ U5 gap: EUR 525m at the 2013 price level → EUR 443m in 2006 prices.
- ▶ Combined construction cost: $C_{2006} \simeq$ **EUR 731m.**

$$\frac{PV_{2006}(B)}{C_{2006}} = \frac{\text{EUR } 3.32\text{bn}}{\text{EUR } 0.731\text{bn}} \simeq 4.5.$$

- ▶ High benefit-cost ratio, but stylized modelling of travel times likely implies overestimation
 - ▶ Even higher when 1.15bn (or 8.77bn in open city) land value incidence is considered

Sources: BVG; Die Zeit; Destatis. Reported totals are price-deflated because annual expenditure profiles are unavailable; operating and maintenance costs are omitted.

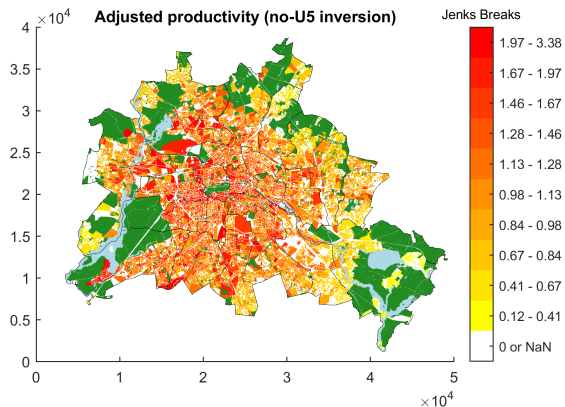
Capstone task

1. Produce and audit the baseline and U5 network scenarios.
2. Compute and validate both bilateral travel-time matrices.
3. Quantify the ARSW model: invert the baseline and solve the U5 counterfactual under closed- and open-city closures.
4. Compute relative changes in local and aggregate outcomes and construct a transparent benefit–cost approximation.
5. Interpret the mechanisms, spatial incidence, welfare effects, and limitations.

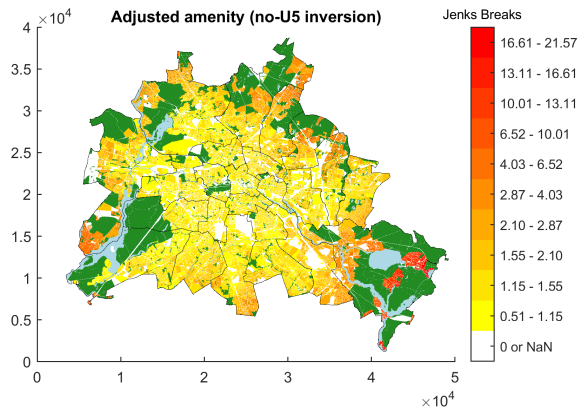
Possible extensions

- ▶ Mechanisms
 - ▶ Direct change in travel times, before relocation.
 - ▶ Equilibrium with exogenous productivity and amenities.
 - ▶ Equilibrium with endogenous productivity and amenity spillovers.
- ▶ Robustness and sensitivity
 - ▶ Test sensitivity to commuting elasticities, agglomeration parameters, etc.
- ▶ Nerd task: Compute richer travel-time matrices
 - ▶ More modes: S-Bahn, bus, tram, car
 - ▶ Transfer, waiting-time
 - ▶ Generalized travel cost

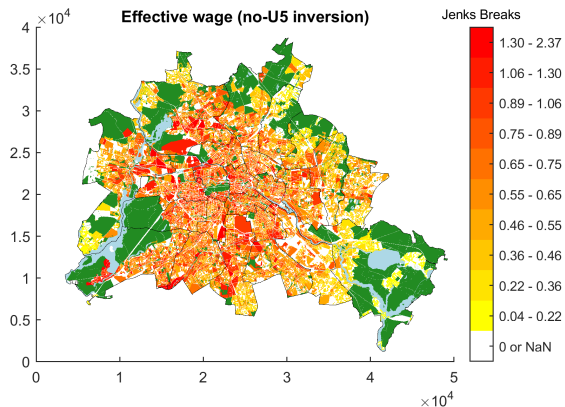
Inverted adjusted productivity

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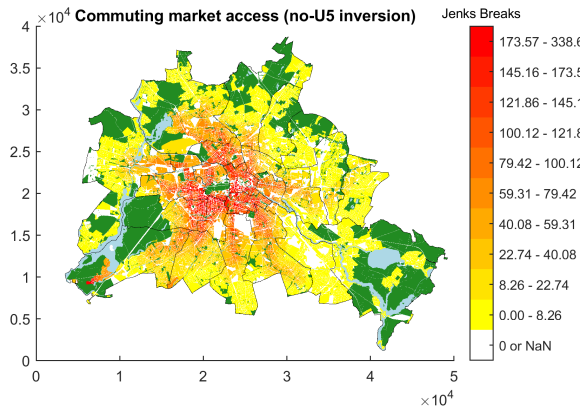
Inverted adjusted amenity

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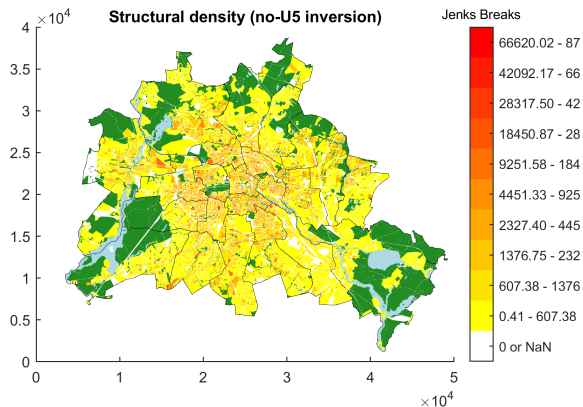
Inverted effective wage

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Inverted commuting market access

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Inverted structural density

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